

A Pedagogical Approach to STEM Education using Mixed Reality

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Abstract

The key to India becoming a developed nation in the next twenty-five years is through technological innovations. These innovations shall lead to product/services development that provide solution to various societal problems indigenously. Technology always fascinate students and it is necessary that they are exposed to it at the right stage. Virtual Reality (VR) and Augmented Reality (AR) are two of the well-known industry 4.0 technologies that have potential to transform the world. Their amalgamation with IoT, 3D printing, AI, Data Analytics etc. will take innovation to a new level. Mixed reality is a technology that encompasses the best of VR and AR and creates a space for learning where digital and physical objects can co-exist and interact immersivity. Mixed reality applications are more effective in visualizations of complex 3D structures that are otherwise difficult to illustrate in 2D environments and phenomena that are invisible to the naked eye. The present work highlights an immersive approach through the use of various open domain visualization and designing applications such as ShapesXR and Gravity Sketch which are deployed on Meta Quest 3 headset to design 3D models and visualize various topics under STEM. An online library with basic templates for STEM models has been developed to facilitate beginners incorporate mixed reality into STEM learning. The efforts showcase the transformative potential of mixed reality technology in STEM education by enhancing the cognitive perceivance of learners.

Introduction

• Mixed Reality:

Mixed Reality (MR) integrates virtual and physical environments, creating immersive experiences where digital content interacts with the real world in real-time. MR seamlessly embeds virtual objects into the user's surroundings, enabling interactive and contextual experiences. MR devices such as Meta Quest 3 utilize sophisticated sensors and spatial mapping to integrate virtual elements seamlessly, facilitating applications in gaming, education, training, design, and beyond.

• Open Source Applications:

ShapesXR



VR design tools for creating 3D models. Sculpt and manipulate objects with ease.

GravitySketch



Sketching and 3D modelling fostering creative teamwork and idea generation

Blender [1]



Open-source 3D creation suite Modelling, Animation, Rendering

• Importance of visual learning in STEM:

Visual learning enhances STEM education by offering immersive, interactive experiences through MR, it enables students to visualize complex concepts, such as molecular structures or astronomical phenomena, biological processes in 3D, fostering deeper understanding and engagement. It facilitates hands-on experimentation and simulation, allowing learners to explore scientific principles in a dynamic environment.

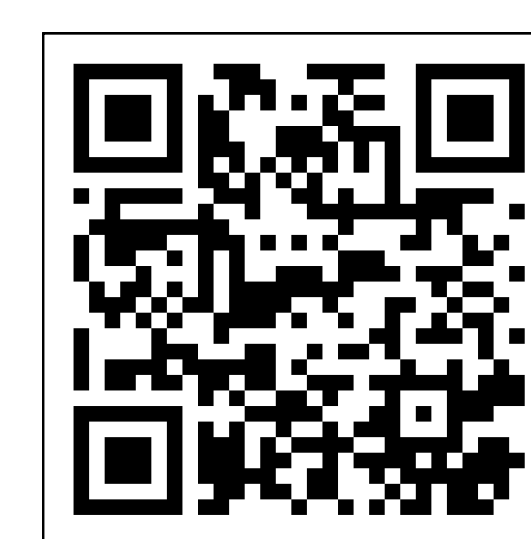
Objective

1. Enhancing the illustrations in STEM education using open-source applications

2. Online library provides basic models for beginners to incorporate MR in learning and understanding of concepts.

3. Provides collaborative space for educator and learner to visualize and explore concepts together.

Prerequisite for a live Demo:



Visit our website

Download 3D viewer application through website

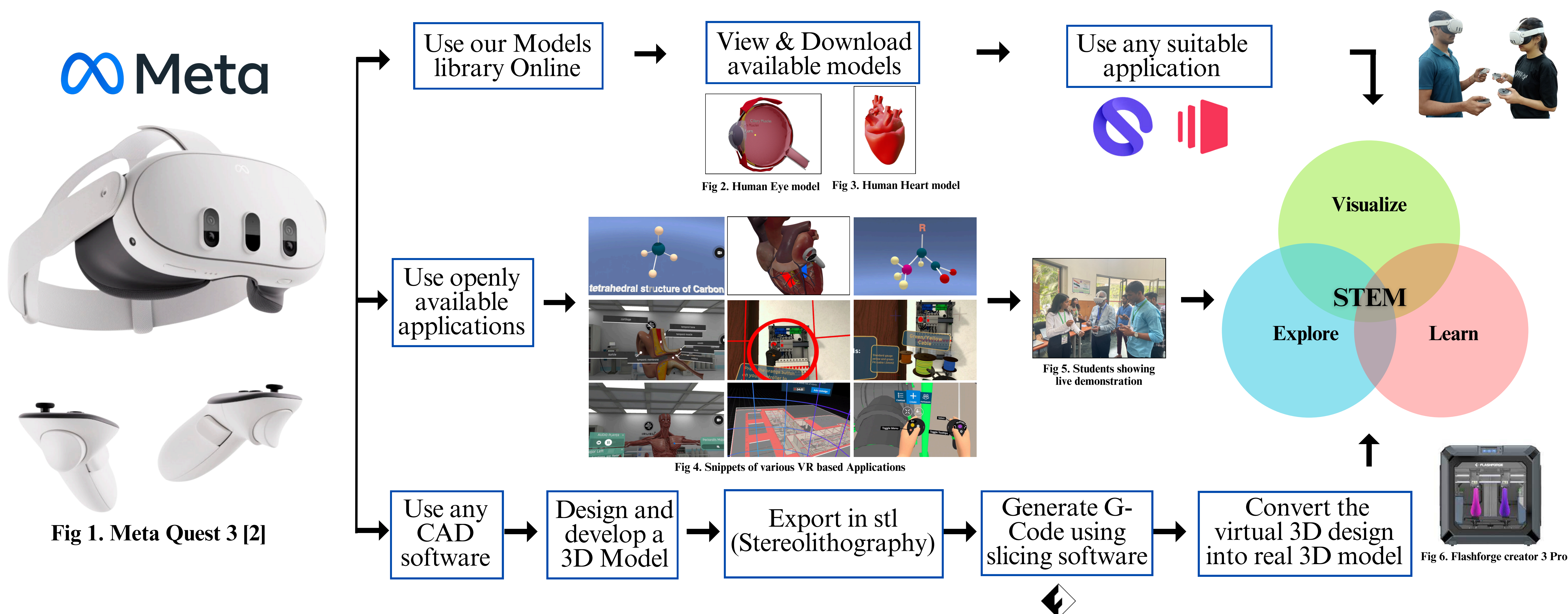
Download the model you want to view from our library

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Time to Experience through Mixed reality!

Methodology



Future Perspect

Simplified 3D designing & modelling process

Laboratory simulation in Mixed Reality

Entrepreneur Mixed Reality integrated teaching platform

Learning by Immersively gamified content

Fig 6. Chart representing future applications of MR in STEM education [3]

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Result & Conclusion

- Using Mixed Reality in STEM education provides a collaborative space to explore the world hidden in plain sight.
- It allows learners to comprehend concepts from the educator's perspective while creatively collaborating and exploring themselves.
- This innovative approach to STEM learning offers an audiovisual learning style, enabling continuous learning outside formal educational settings.
- It is more engaging and memorable than traditional text-based learning, leading to better comprehension and retention of information.

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